



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

SCORE - II

19-05-2021

JEE(Main + Advance) : Enthusiast & Leader Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Main)

TEST DATE : 19-05-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	B	B	A	A	A	A	C
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	C	A	D	B	A	A	A	C

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	C	A	C	A	B	B	C	D
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	B	C	B	C	C	C	D	B

MATHEMATICS

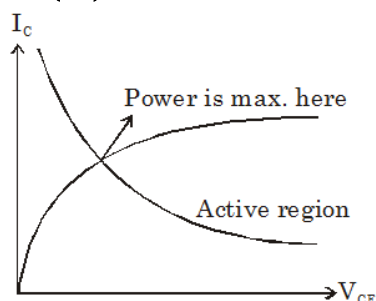
SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	C	C	C	B	A	C	D
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	D	B	D	B	D	D	C	C

HINT – SHEET

PHYSICS

SEC-A

1. Ans (C)



2. Ans (C)

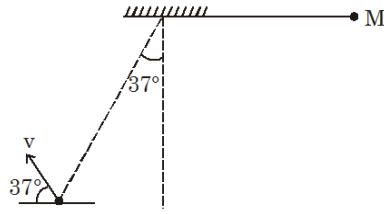
$$WE = \Delta K$$

WD_{friction} on a system is independent of r-f

$$\therefore -\mu mg\ell = \frac{1}{2}\mu v_{if}^2 - \frac{1}{2}\mu v_{fi}^2$$

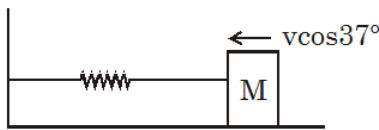
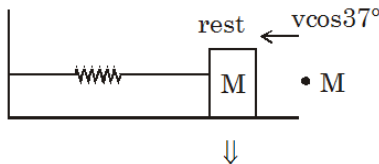
$$\Rightarrow -\mu mg\ell = -\frac{1}{2} \frac{mM}{M+m} (u-0)^2$$

3. Ans (A)



$$mg \ell \cos 37^\circ = \frac{1}{2}mv^2$$

$$\therefore v = \sqrt{20}$$



$$\frac{1}{2}m(v \cos 37^\circ)^2 = \frac{1}{2}kx^2$$

4. Ans (B)

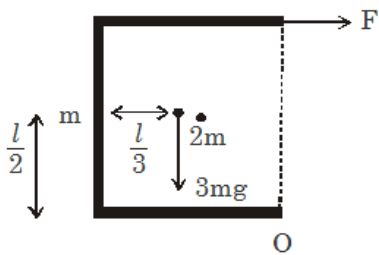
COM of system does not move as $F_{\text{net ext}} = 0$

$$x = 0$$

$$\therefore 0 = m_1 \Delta x_{1G} + m_2 \Delta x_{2G} + M \Delta x_{MG}$$

$$0 = 50 [50 + \Delta x] + 60 [-50 + \Delta x] + 40 \Delta x$$

5. Ans (B)



$$C_0 = 0 \Rightarrow F \times \ell - 3mg \frac{2\ell}{3} = 0$$

$$F = 2mg$$

6. Ans (A)

5 antinodes $\Rightarrow$ 5 loops are formed

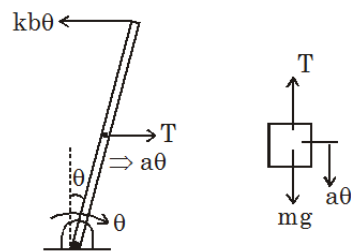
$$f = \frac{5}{2\ell} \sqrt{\frac{9g}{\mu}} \dots (1)$$

Same $t-f \Rightarrow$ same 'f'

$$f = \frac{3}{2\ell} \sqrt{\frac{Mg}{\mu}} \dots (2)$$

$$\therefore M = 5 \text{ kg}$$

7. Ans (A)



$$T \cos \theta - kb \theta \cdot b \cos \theta = 0 \dots (1)$$

$$mg - T = \frac{md^2(a\theta)}{dt^2} = 0 \dots (2)$$

$$\Rightarrow mg - \frac{kb^2}{a} \theta = ma \frac{d^2 \theta}{dt^2}$$

$$\therefore \frac{d^2 \theta}{dt^2} = \frac{g}{a} - \frac{kb^2}{ma^2} \theta$$

$$\therefore \omega^2 = \sqrt{\frac{k}{m}} \frac{b}{a}$$

8. **Ans (A)**

Let V be the initial volume of the gas. It is expanded to a volume ηV . The workdone in this process is given by

$$W = \int_V^{\eta V} P dV = \int_V^{\eta V} \alpha V dV = \alpha \left[\frac{V^2}{2} \right]_V^{\eta V}$$

$$W = \frac{\alpha V^2}{2} [\eta^2 - 1]$$

The pressure of the gas varies with volume $P = \alpha V$.

So, the initial and final pressures will be αV and $\eta \alpha V$. The change in internal energy is given by

$$U = \frac{P_f V_f - P_i V_i}{\gamma - 1} = \frac{\eta \alpha V^2 - \alpha V^2}{\gamma - 1}$$

$$= \frac{\alpha V^2}{\gamma - 1} [\eta^2 - 1]$$

The heat exchange in this process is given by

$$Q = U + W$$

$$= \frac{\alpha V^2}{\gamma - 1} [\eta^2 - 1] + \frac{\alpha V^2}{2} [\eta^2 - 1]$$

$$= \frac{\alpha V^2}{2} [\eta^2 - 1] \left[\frac{2}{\gamma - 1} + 1 \right]$$

$$= \frac{\alpha V^2}{2} [\eta^2 - 1] \left[\frac{\gamma + 1}{\gamma - 1} \right]$$

$$\text{Here, } T_i = \frac{P_i V_i}{nR} = \frac{\alpha V^2}{nR}$$

$$\text{and } T_f = \frac{P_f V_f}{nR} = \frac{\eta^2 \alpha V^2}{nR}$$

$$\text{Now, Heat capacity } C = \frac{Q}{T_f - T_i}$$

$$\therefore C = \frac{1}{T_f - T_i} \left[\frac{\alpha V^2}{2} (\eta^2 - 1) \left\{ \frac{\gamma + 1}{\gamma - 1} \right\} \right]$$

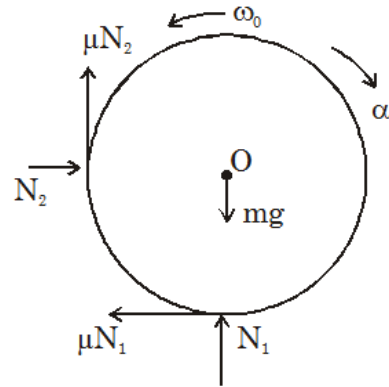
$$= \frac{nR}{\alpha V^2 (\eta^2 - 1)} \times \left[\frac{\alpha V^2}{2} (\eta^2 - 1) \left\{ \frac{\gamma + 1}{\gamma - 1} \right\} \right]$$

$$= \frac{nR}{2} \left\{ \frac{\gamma + 1}{\gamma - 1} \right\}$$

Here, $n = 1$

$$\therefore C = \frac{R}{2} \left(\frac{\gamma + 1}{\gamma - 1} \right)$$

9. **Ans (A)**



$$N_1 + \mu N_2 = mg \quad \dots (1)$$

$$N_2 = \mu N_1 \quad \dots (2)$$

$$\Rightarrow N_1 = \frac{mg}{1 + \mu^2}$$

$$N_2 = \frac{\mu mg}{1 + \mu^2}$$

$$\tau_0 : \mu N_1 R + \mu N_2 R = I \alpha = \frac{1}{2} m R^2 \alpha \quad \dots (3)$$

$$\omega^2 - \omega_0^2 = 2 \alpha \theta \quad \dots (4)$$

$$\Rightarrow 0 - \omega_0^2 = 2 (-\alpha) \theta$$

$$\therefore \eta = \frac{\theta}{2\pi}$$

10. **Ans (C)**

$$\frac{1}{k_{eq}} = \frac{1}{k} + \frac{1}{2k} + \frac{1}{2^2 k} + \frac{1}{2^3 k} + \dots + \frac{1}{2}$$

$$= \frac{1}{k} \frac{1}{(1 - \frac{1}{2})} = \frac{2}{k}$$

$$k_{eq} = \frac{k}{2}$$

$$T = 2\pi \sqrt{\frac{2m}{k}}$$

11. **Ans (B)**

$$v = \sqrt{gR \tan \theta} : \text{for no friction to act}$$

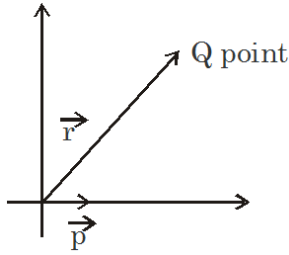
$$= \sqrt{10 \times 30 \times \frac{3}{4}} = 15 \text{ m/s}$$

The speed of car is less than 15

$\therefore$ it takes to slip down

$\therefore$ friction acts up the incline

12. Ans (B)



$$|\vec{E}_p| = |\vec{V}_p|$$

$$\theta = 45^\circ$$

$$\frac{kP}{r^3} \sqrt{1 + 3\sin^2\theta} = \frac{kP \sin\theta}{r^2}$$

angle between $\vec{r}$ & $\vec{p}$

$$\frac{\sqrt{1 + 3\sin^2\theta}}{r} = \sin\theta$$

$$\frac{\sqrt{1 + 3\sin^2\theta}}{\sqrt{5}} = \sin\theta$$

sq. both side

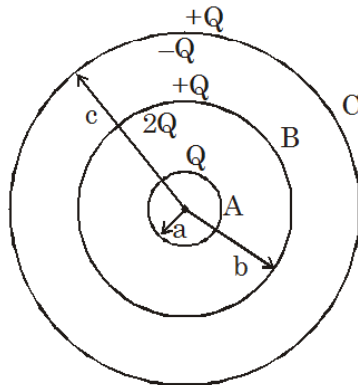
$$1 + 3\sin^2\theta = 5\sin^2\theta$$

$$1 = 2\sin^2\theta$$

$$\sin\theta = \frac{1}{2}$$

$$\text{So, } \vec{r} = \frac{\sqrt{5}}{2} (\hat{i} + \hat{j})$$

13. Ans (C)



(A) obviously same charge

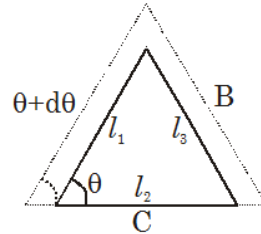
$$(B) \sigma_{B-\text{inner}} = -\frac{Q}{4\pi b^2}$$

$$\sigma_{C-\text{inner}} = -\frac{Q}{4\pi c^2}$$

(C) true

14. Ans (A)

When the system is heated, the rods expand and the triangle does not remain equilateral. Let lengths of rods A, B and C be ℓ_1 , ℓ_2 and ℓ_3 respectively then



$$\cos\theta = \frac{l_1^2 + l_2^2 - l_3^2}{2l_1l_2}$$

$$\text{or } 2l_1l_2 \cos\theta = l_1^2 + l_2^2 - l_3^2 \quad \dots(i)$$

Differentiating Eq.(i), we get

$$2l_1 \cos\theta dl_2 + 2l_2 \cos\theta dl_1 - 2l_1l_2 \sin\theta d\theta$$

$$= 2l_1 dl_2 + 2l_2 dl_1 - 2l_3 dl_3 \quad \dots(ii)$$

Let the temperature of the system be increased

by ΔT . Then

$$\left. \begin{aligned} dl_1 &= l_1 \alpha_1 \Delta T \\ \text{and } dl_2 &= l_2 \alpha_1 \Delta T \\ dl_3 &= l_3 \alpha_2 \Delta T \end{aligned} \right\} \quad \dots(iii)$$

In case of equilateral triangle $\ell_1 = \ell_2 = \ell_3$

and $\theta = 60^\circ$ (say)

From Eqs. (ii) and (iii), we get :

$$2l^2 \cos 60^\circ \alpha_1 \Delta T + 2l^2 \cos 60^\circ \alpha_1 \Delta T$$

$$- 2l^2 \sin 60^\circ d\theta$$

$$= 2l^2 \alpha_1 \Delta T + 2l^2 \alpha_1 \Delta T - 2l^2 \alpha_2 \Delta T$$

$$\frac{\alpha_1}{2} \Delta T + \frac{\alpha_1}{2} \Delta T - \frac{\sqrt{3}}{2} d\theta$$

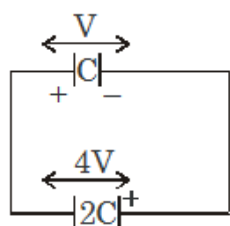
$$= \alpha_1 \Delta T + \alpha_1 \Delta T - \alpha_2 \Delta T$$

$$\alpha_1 \Delta T - \frac{\sqrt{3}}{2} d\theta = 2\alpha_1 \Delta T - \alpha_2 \Delta T$$

$$(\alpha_2 - \alpha_1) \Delta T = \frac{\sqrt{3}}{2} d\theta$$

$$\Delta T = \frac{\sqrt{3} d\theta}{2(\alpha_2 - \alpha_1)}$$

15. Ans (D)



$$|\Delta V| = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 + V_2)^2$$

(for opposite polarity)

$$= \frac{1}{2} \frac{C \times 2C}{3C} (5V)^2$$

$$= \frac{25CV^2}{3}$$

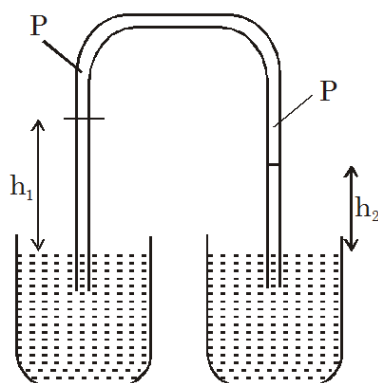
16. Ans (B)

$$\frac{\mu_0 J r}{2} = \frac{\mu_0 i}{2\pi R}$$

$$\Rightarrow \frac{i}{\pi a^2} \cdot r = \frac{i}{\pi R}$$

$$\therefore a = \sqrt{rR}$$

17. Ans (A)

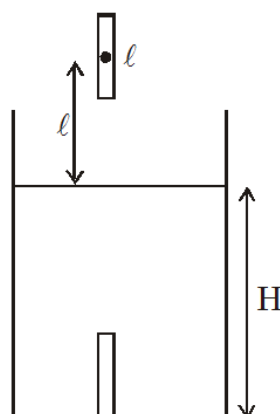


$$P_0 = P + \rho_1 g h_1 \quad \dots (1)$$

$$P_0 = P + \rho_2 g h_2 \quad \dots (2)$$

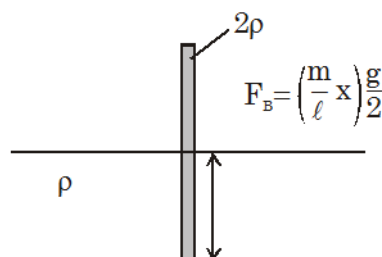
$$\therefore \rho_2 = \rho_1 \frac{h_1}{h_2} = 28$$

18. Ans (A)



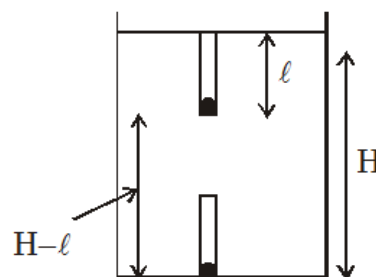
$$WD = \Delta K$$

$$mg \left[h - \frac{\ell}{2} + H \right] + WD_B = \frac{1}{2} mv^2$$



$$WD_{b-1} = \int_0^{\ell} \left(\frac{mgx}{2\ell} \right) dx = -\frac{mg\ell}{4}$$

$$WD_{b-2} = -\frac{mg}{2} (H - \ell) : \text{when completely inside water}$$



$$\therefore mg \left[h - \frac{\ell}{2} + H \right] - \frac{mg\ell}{4} - \frac{mg(H - \ell)}{2}$$

$$= \frac{1}{2} mv^2$$

19. Ans (A)

The mean energy of electrons

$$\langle E \rangle = \frac{\int_0^{E_{\max}} E n(E) dE}{\int_0^{E_{\max}} n(E) dE}$$

$$\text{Given } n(E) dE = c \sqrt{E} (E_{\max} - E)^2 dE$$

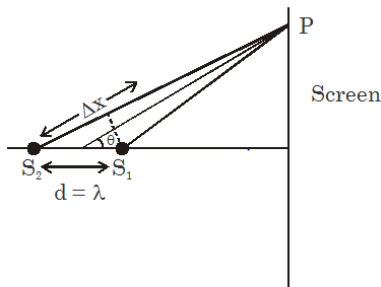
where $c = \text{constant}$

$$\langle E \rangle = \frac{c \int_0^{E_{\max}} E \sqrt{E} (E_{\max} - E)^2 dE}{c \int_0^{E_{\max}} \sqrt{E} (E_{\max} - E)^2 dE} = \frac{E_{\max}}{3}$$

If all the electrons emitted are absorbed then the kinetic energy of the electrons is converted into heat.

$$\begin{aligned} \text{Heat evolved/sec} &= (\text{mean energy})(\text{no. of electrons emitted/second}) \\ &= (0.156) \times (3.7 \times 10^7) / 3 \text{ MeV/s} \\ &= 1.92 \times 10^6 \text{ MeV/s} \end{aligned}$$

20. Ans (C)



$$\Delta x = S_2P$$

$$\Delta x = d \cos \theta$$

$$\Delta x = \lambda \cos \theta$$

the min path digs (at $\theta = 90^\circ$)

$$\Delta x = \lambda \cos 90^\circ$$

$$(\Delta x_{\min} = 0)$$

the max path digs (at $\theta = 0^\circ$)

$$\Delta x = \lambda \cos 0^\circ$$

$$(\Delta x_{\max} = \lambda)$$

thus in between these two positions there is only 1 min for which

$$\Delta x = \frac{\lambda}{2} \text{ So, } \frac{\lambda}{2} = \lambda \cos \theta$$

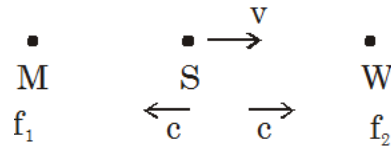
$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

PHYSICS

SEC-B

1. Ans (3.00)



$$f_1 = \frac{c}{c + v} f$$

$$f_2 = \frac{c}{c - v} f$$

same f_2 is received by man

$$\therefore f = f_2 - f_1 = \frac{2cv}{c^2 - v^2} f$$

$$\simeq \frac{2cv}{c^2} f$$

$$= \frac{2v}{c} f$$

$$3 = \frac{2 \times v \times 175}{350}$$

$$\therefore v = 3 \text{ m/s}$$

2. Ans (5.00)

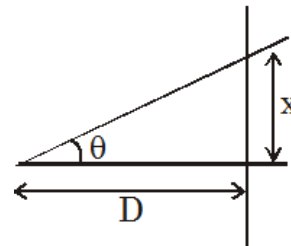
$$-KA \frac{dT}{dx} = 6000$$

$$\Rightarrow K = \frac{6000}{(1-x) 300 (4+3x^2)}$$

at $x = 0$

$$K = 5.$$

3. Ans (1.08 to 1.09)



$$d \sin \theta = n \lambda$$

$$\Rightarrow \sin \theta \simeq \theta = \frac{\ell}{d}$$

$$\tan \theta = \frac{x}{D} \simeq \theta$$

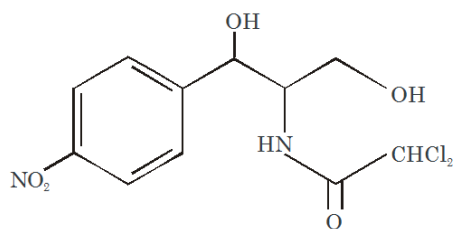
$$\frac{x}{D} = \frac{\lambda}{d} \therefore D = \frac{xd}{\lambda}$$

4. Ans (2.60)

$$M = \frac{26}{1} \frac{25}{2.5} = 2.60$$

3. Ans (C)

Chloramphenicol



4. Ans (A)

$$\Delta H^0 = -10,500, \Delta n_g = 0$$

$$\Delta H^0 = \Delta H^0 - T\Delta S^0$$

$$= -10,500 - 300 \times (-50) + 5000 \text{ J}$$

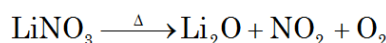
$$\therefore \Delta G^0 = 5 \text{ kJ, reaction is non-spontaneous}$$

5. Ans (C)

1 mole $\text{CoCl}_3 \cdot 5\text{NH}_3$ will give 2 moles of AgCl

$\therefore$ 2 chlorine atoms will satisfy the only primary valency.

8. Ans (B)



9. Ans (C)

Except glycine, all other naturally occurring α -amino acids are optically active

10. Ans (D)

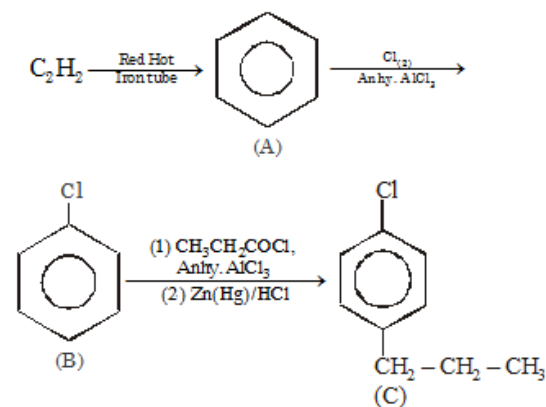
95% ethanol & 5% water by volume is azeotropic composition.

11. Ans (A)

Mischmetal is an alloy which mostly consists of Lanthanoids metals (~95%)

Ce^{+4} is colourless

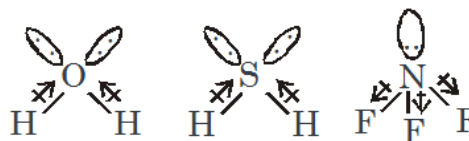
12. Ans (B)



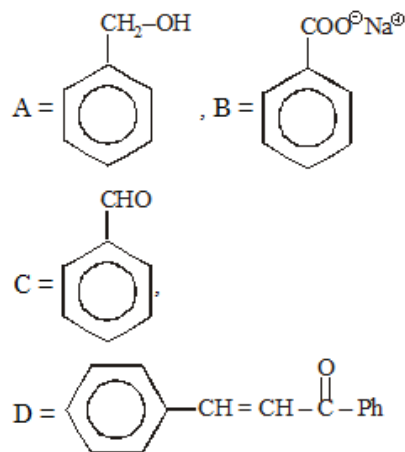
13. Ans (B)

$$\text{Rate } K = [\text{H}_2\text{O}_2][\text{I}^-]$$

14. Ans (C)



15. Ans (B)

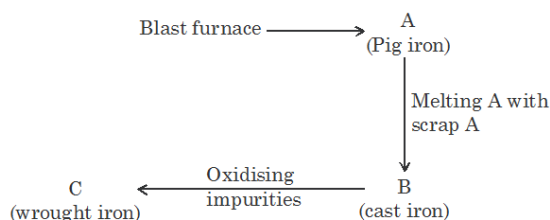


16. Ans (C)

Correct order of V_C for the given gases is

$$V_C(\text{N}_2) > V_C(\text{H}_2) > V_C(\text{He})$$

17. Ans (C)



18. Ans (C)

Industrially ethyne is produced by the hydrolysis of calcium carbide.

19. Ans (D)

$$\text{Radial nodes} \rightarrow (n - l - 1)$$

For 1s-orbital, the probability density is max at nucleus.

20. Ans (B)

(A) In ice crystal, each oxygen atom is

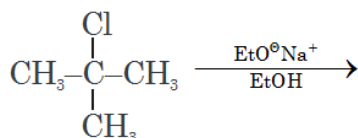
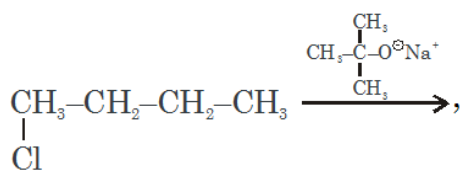
surrounded tetrahedrally by 4 other oxygens.

(C) 10V H_2O_2 contains 33% H_2O_2

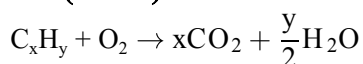
CHEMISTRY

SEC-B

1. Ans (2.00)



2. Ans (26.00)



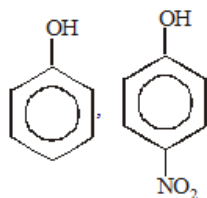
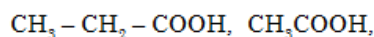
$$\text{M.F.} = \text{C}_2\text{H}_2$$

$$\text{M.wt} = 26$$

3. Ans (7.00)

Except Ba^{+2} , Sr^{+2} all other cations are precipitated by NH_4OH

4. Ans (4.00)

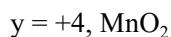
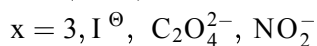


5. Ans (15.00)

$$\text{Ionization constant} : 8 \times 10^{-7}$$

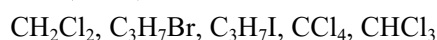
$$a = 8, b = 7, a + b = 15$$

6. Ans (7.00)



$$z = 0$$

7. Ans (5.00)



9. Ans (6.00)

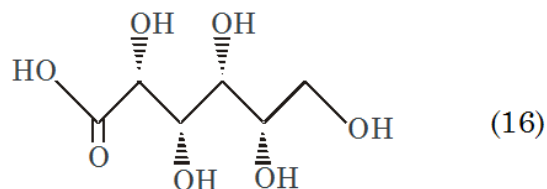
$$\text{B}_2 \rightarrow 0$$

$$\text{N}_2 \rightarrow 4$$

$$\text{C}_2 \rightarrow 2$$

$$0 + 2 + 4 = 6$$

10. Ans (16.00)



MATHEMATICS

SEC-A

1. Ans (C)

$$\sin x \cos x [\sin^2 x + \sin x \cos x + \cos^2 x] = 1$$

$$\Rightarrow \sin x \cos x + (\sin x \cos x)^2 = 1$$

$$\sin^2 2x + 2 \sin 2x - 4 = 0 \Rightarrow \sin 2x$$

$$= \frac{-2 \pm \sqrt{4 + 16}}{2}$$

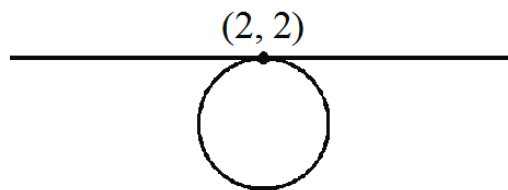
$$= -1 \pm \sqrt{5}, \text{ which is not possible.}$$

2. Ans (B)

Equation of tangent at (2, 2)

$$2x + \left(\frac{x+2}{2}\right) - \left(\frac{y+2}{2}\right) + \left(\frac{2x+2y}{2}\right) = 8$$

$$4x + x + 2 - y - 2 + 2x + 2y = 16$$



$$7x + y - 16 = 0$$

$$(x-2)^2 + (y-2)^2 + \lambda(7x + y - 16) = 0$$

$$x^2 + y^2 + x(7\lambda - 4) + y(\lambda - 4) + 8 - 16\lambda = 0$$

$$\text{which also touches the line } x - 7y + 37 = 0$$

Applying condition of tangency, get

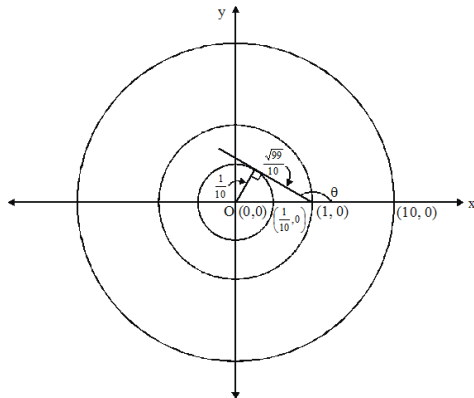
$$x^2 + y^2 + 3x - 3y - 8 = 0$$

or

$$x^2 + y^2 - 11x - 5y + 24 = 0$$

$$\therefore |a + b + c| = 8.$$

3. Ans (C)



Put $\frac{z_1}{z_2} = z$

$\therefore \left| \frac{z_1}{z_2} \right| = |z| \Rightarrow |z| = \frac{1}{10}$

Also, $\theta = \arg \left(\frac{z_1}{z_2} - 1 \right) = \arg(z - 1)$

$\therefore \tan^2 \theta \Big]_{\max.} = \frac{1/100}{99/100} = \frac{1}{99}$

4. Ans (C)

$\sum_{m=0}^{\infty} (\alpha^m + \beta^m) = (1 + \alpha + \alpha^2 + \dots) + (1 + \beta + \beta^2 + \dots)$

$= \frac{1}{1-\alpha} + \frac{1}{1-\beta} = \frac{2 - (\alpha + \beta)}{1 - (\alpha + \beta) + \alpha\beta} = 6$

5. Ans (C)

Required Words = Total - {n(G) + n(S) - n(G ∩ S)}

(where n(G) = words start with G &

n(S) = words end with S)

$n(G) = n(S) = \frac{720}{6} = 120$

$n(G \cap S) = G_ _ _ _ S = 24$

Required Words = 720 - (120 + 120 - 24) = 504

6. Ans (C)

$n(S) = 12^6$

Now, any two months can be chosen

in $^{12}C_2$ ways. The six birthday can fall in these two months in 2^6 ways. Out of these 2^6 ways there are two ways when all the six birthdays fall in one month so favourable number of ways in $^{12}C_2 \times (2^6 - 2)$.

Hence required probability is

$= \frac{^{12}C_2 \times (2^6 - 2)}{12^6} = \frac{12 \times 11 \times (2^6 - 2)}{2 \times 12^6}$

$\frac{341}{12^5}$

7. Ans (B)

Put $\begin{vmatrix} (1-\lambda) & 3 & -4 \\ 1 & -(3+\lambda) & 5 \\ 3 & 1 & -\lambda \end{vmatrix} = 0$

$\Rightarrow \lambda(\lambda + 1)^2 = 0 \Rightarrow \lambda = -1, 0$

8. Ans (A)

p	q	$p \rightarrow q$	$\sim p$	$q \vee \sim p$	$(p \rightarrow q) \leftrightarrow (q \vee \sim p)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

9. Ans (C)

$S = \sum_{m=1}^{12} 12m \cdot \frac{12}{m} {}^{11}C_{m-1} \cdot {}^{11}C_{m-1}$

$= 12^2 \sum_{m=1}^{12} ({}^{11}C_{m-1})^2 = 12^2 \cdot {}^{22}C_{11}$

$\Rightarrow p = 2 \text{ \& } q = 11 \Rightarrow p + q = 13$

10. Ans (D)

$y = f(x) \Rightarrow f(1) = 7 \Rightarrow y = 7 \text{ when } x = 1$

$x = g(y)$ (where g is the inverse of f)

$1 = g'(y) \cdot f'(x)$

$\Rightarrow g''(y) = -\frac{f''(x)}{[f'(x)]^3}$ (put $x = 1, y = 7$)

$\Rightarrow g''(7) = -\frac{f''(1)}{[f'(1)]^3} = -\frac{16}{(2)^3} = -2$

11. Ans (D)

Given Limit can be written as

$\lim_{t \rightarrow 0} \frac{(3^t - 1)(4^t - 1)(\sqrt{2 \cos t + 7} + 3)}{2 \cos t - 2}$
 $= \lim_{t \rightarrow 0} \frac{(3^t - 1)}{t} \cdot \frac{(4^t - 1)}{t} \cdot \frac{(\sqrt{2 \cos t + 7} + 3)}{-2 \frac{(1 - \cos t)}{t^2} t^2}$

$= \log_e 3 \cdot \log_e 4 \cdot (-6) = -6 \log_e 3 \cdot \log_e 4$

12. Ans (C)

$$f(x) = \begin{cases} x^2 + ax + b; & x < -1 \\ 2x & -1 \leq x \leq 1 \\ x^2 + ax + b; & x > 1 \end{cases}$$

f is continuous at $x = 1$

$$f(1^+) = f(1^-) = f(1)$$

$$1 + a + b = 2 \Rightarrow a + b = 1 \dots\dots(i)$$

f is continuous at $x = -1$

$$f(-1^+) = f(-1) = f(1^-)$$

$$-2 = 1 - a + b \Rightarrow a - b = 3 \dots\dots(ii)$$

from (i) & (ii); $a = 2, b = -1$

$$\text{Now } f(x) = \begin{cases} x^2 + 2x - 1; & x < -1 \\ 2x & -1 \leq x \leq 1 \\ x^2 + 2x - 1; & x > 1 \end{cases}$$

$$f'(1^+) = 2 + 2 = 4; f'(1^-) = 2 \text{ Non-diff. at } x = 1$$

$$f'(-1^+) = 2; f'(-1^-) = -2 + 2 = 0 \text{ Non-diff. at } x = -1$$

13. Ans (D)

In $x, x+1, e^x - e^2$ becomes zero at $x = 1, -1$ and 2 respectively, for which $\sin px$ is also zero

Hence $f(x)$ is derivable at these points.

$\therefore$ number of points of non derivability = 0

14. Ans (B)

$$f(x) = ax^3 + bx^2 + cx + 4$$

$$\text{solving with tangent } y = \frac{5x}{3} + \frac{100}{27}$$

$$ax^3 + bx^2 + cx + 4 - \left(\frac{5x}{3} + \frac{100}{27} \right) = K$$

$$\left(x + \frac{2}{3} \right)^3 \dots\dots\dots(i)$$

put $x = 0$

$$4 - \frac{100}{27} = K \times \frac{8}{27} \Rightarrow K = 1$$

$$\text{put } x = 1 \quad a + b + c + 4 - \frac{145}{27} = \frac{125}{27}$$

$$a + b + c = 6$$

15. Ans (D)

$$\text{Let } F(x) = f^3(x)$$

and $F(x)$ is continuous and differentiable function in $[1, 3]$.

$$\therefore \frac{F(3) - F(1)}{3 - 1} = F'(c) \text{ (using L.M.V.T.)}$$

$$\frac{f^3(3) - f^3(1)}{2} = 3f(2(c)) \cdot f'(c) \Rightarrow k = 6$$

16. Ans (B)

$$I = \int \left[1 + \frac{(x-1)(x-2)(x-3) - (x-4)(x-5)(x-6)}{(x-4)(x-5)(x-6)} \right] dx$$

$$= \int \left[1 + \frac{3 \times 2 \times 1}{(x-4)(-1)(-2)} + \frac{4 \times 3 \times 2}{1(x-5)(-1)} + \frac{5 \times 4 \times 3}{2(1)(x-6)} \right] dx$$

$$= x + 3 \ln|x-4| - 24 \ln|x-5| + 30 \ln|x-6| + c$$

17. Ans (D)

$$I_1 = \frac{b-a}{2} = \frac{\pi/2 - 0}{2}$$

$$I_1 = \frac{\pi}{4}$$

$$I_2 = \int_{-\pi}^{\pi} \frac{\cos^2(x)}{1+a^x} dx$$

apply King and add

$$2I_2 = \int_{-\pi}^{\pi} \cos^2(x) \cdot dx = 2 \int_0^{\pi} \cos^2(x) \cdot dx$$

$$\Rightarrow I_2 = \int_0^{\pi} \cos^2 x \, dx = \frac{\pi}{2}$$

18. Ans (D)

$$x^2 < x^{\frac{\pi}{2}} < x \quad \forall x (0, 1)$$

$$\Rightarrow \frac{1}{1+x} < \frac{1}{1+x^{\frac{\pi}{2}}} < \frac{1}{1+x^2}$$

$\Rightarrow$

$$(\ln(1+x))_0^1 < \int_0^1 \frac{1}{1+x^{\frac{\pi}{2}}} dx < (\tan^{-1}x)_0^1$$

$$\Rightarrow \log_e 2 < I < \frac{\pi}{4}$$

19. Ans (C)

$$\sqrt{x} + \sqrt{|y|} = 1$$

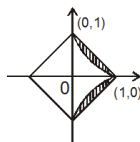
Above curve is symmetric about x-axis

$$\sqrt{|y|} = 1 - \sqrt{x} \text{ and } \sqrt{x} = 1 - \sqrt{|y|}$$

$$\Rightarrow \text{for } x > 0, y > 0 \quad \sqrt{y} = 1 - \sqrt{x}$$

$$\frac{1}{2\sqrt{y}} \frac{dy}{dx} = -\frac{1}{2\sqrt{x}}$$

$$\frac{dy}{dx} = -\sqrt{\frac{y}{x}} < 0,$$



function is decreasing required area

$$= 2 \int_0^1 (2\sqrt{x} - 2x) = \frac{1}{3} = \frac{2}{3}$$

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20. Ans (C)

$$\frac{dy}{dx} = -\frac{x(x^2 + 3y^2)}{y(y^2 + 3x^2)} = \frac{-1}{(y/x)} \left(\frac{1 + 3(y^2/x^2)}{(y^2/x^2) + 3} \right)$$

Now put $y = tx$

Aliter :

$$x^3 dx + y^3 dy + \frac{3}{2} [2xy^2 dx + 2x^2y dy] = 0$$

$$\Rightarrow x^3 dx + y^3 dy + \frac{3}{2} d(x^2 y^2) = 0$$

$$\Rightarrow \frac{x^4}{4} + \frac{y^4}{4} + \frac{3}{2} (x^2 y^2) = c' \Rightarrow x^4 + y^4 + 6x^2 y^2 = c$$

MATHEMATICS

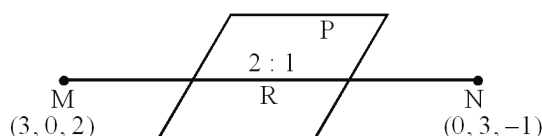
SEC-B

1. Ans (7.00)

$$\text{Let } P_1 : x + y + z + 3 = 0; P_2 : x - y + z - 2 = 0$$

$$\text{Let } P : P_1 + \lambda P_2 = 0$$

$$\Rightarrow (x + y + z + 3) + \lambda(x - y + z - 2) = 0 \dots (1)$$



Put $(x = 1, y = 2, z = 0)$ in equation (1), we

get $\lambda = 2$

$$\therefore P : 3x - y + 3z = 1 \text{ Ans.}$$

$$\Rightarrow (a + b + c) = 3 + 3 + 1 = 7$$

3. Ans (3.00)

$$\text{Let } \vec{a} \wedge \vec{b} = \theta$$

$$\text{Now } \vec{x} = \vec{a} - (\vec{a} \cdot \vec{b})\vec{b}$$

$$\Rightarrow |\vec{x}|^2 = \vec{a}^2 + (\vec{a} \cdot \vec{b})^2 b^2 - 2(\vec{a} \cdot \vec{b})^2$$

$$\Rightarrow |\vec{x}|^2 = 1 + \cos^2 \theta - 2\cos^2 \theta$$

$$\Rightarrow |\vec{x}| = \sin \theta$$

$$\text{Also } \therefore \vec{y} = \vec{a} \times \vec{b} \Rightarrow |\vec{y}| = |\vec{a} \times \vec{b}| = \sin \theta$$

$$\therefore |\vec{x}| = |\vec{y}|$$

$$\text{Also, } \vec{x} \cdot \vec{b} = \vec{a} \cdot \vec{b} - (\vec{a} \cdot \vec{b})b^2 = 0$$

$$\therefore |\vec{x} \cdot \vec{b}| = 0$$

$$\text{Also, } \vec{x} \cdot \vec{a} = a^2 - (\vec{a} \cdot \vec{b})^2 = 1 - \cos^2 \theta = \sin^2 \theta$$

$$\therefore \vec{x} \cdot \vec{a} > 0$$

4. Ans (25.00)

Let $x_1, x_2, x_3, \dots, x_9$ are 9 items they

$$\Rightarrow \frac{x_1 + x_2 + x_3 + \dots + x_9}{9} = 15$$

$$\therefore \sum_{i=1}^9 x_i = 135$$

if one more item x_{10} is added then

$$\sum_{i=1}^{10} x_i = 135 + x_{10}$$

$$\therefore \text{mean } \bar{x} = \frac{135 + x_{10}}{10} = 16$$

$$\therefore x_{10} = 160 - 135 = 25$$

5. Ans (220.00)

$$A = \{1, 2, 3, 4, 5, 6\} \Rightarrow n(A) = 6$$

$$B = \{1, 2\} \Rightarrow n(B) = 2$$

$$\therefore A \cap B = \{1, 2\} \Rightarrow n(A \cap B) = 2$$

$$\text{So, number of elements in } A \times (A \cap B) = 12$$

$$\therefore \text{Number of subsets containing 3 elements} =$$

$${}^{12}C_3 = 220$$

HS-12/13

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2. Ans (9.00)

$$\frac{1}{6} [\vec{a} \vec{b} \vec{c}] = \frac{1}{2} \Rightarrow [\vec{a} \vec{b} \vec{c}] = 3$$

$$\therefore \frac{1}{6} [2 (\vec{a} \times \vec{b}) \quad 3 (\vec{b} \times \vec{c}) \quad (\vec{c} \times \vec{a})]$$

$$= [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] = [\vec{a} \vec{b} \vec{c}]^2 = 9.$$



6. Ans (11.00)

$$M = \text{adj. } N$$

$$\Rightarrow |M| = |\text{adj. } N| = |N|^2$$

$$\therefore |M| = (4)^2 \Rightarrow |M| = 16$$

$$\Rightarrow \begin{vmatrix} 1 & c & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{vmatrix} = 16$$

$$\Rightarrow c = 11$$

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7. Ans (16.00)

$$e_1^2 = 1 - \frac{2}{3} = \frac{1}{3} \Rightarrow e_1 = \frac{1}{\sqrt{3}}$$

$$\text{As, } e_1 e_2 = \frac{1}{2} \Rightarrow e_2 = \frac{\sqrt{3}}{2}$$

$$\text{Now, } \frac{3}{4} = 1 - \frac{16}{b^2} \Rightarrow b = 8$$

8. Ans (2.00)

$a_1, a_2, a_3, \dots$ are in A.P.

$$d = 2, a_{10} = 21 \Rightarrow a_1 = 3, a_n = 2n + 1$$

$$\sum_{r=1}^{\infty} b_r = \sum_{n=1}^{\infty} \frac{1}{(2n+1)(2n+5)}$$

$$= \frac{1}{4} \sum_{n=1}^{\infty} \left(\frac{1}{2n+1} - \frac{1}{2n+5} \right) = \frac{1}{4} \left[\frac{1}{3} + \frac{1}{5} \right]$$

$$= \frac{2}{15}$$

9. Ans (2.00)

$$z^2 + z = i\bar{z}$$

$$z(z+1) = i\bar{z}$$

$$|z| |z+1| = |z|$$

$$\Rightarrow |z+1| = 1 (\because z \neq 0)$$

$$\Rightarrow (z+1)(\bar{z}+1) = 1$$

$$\Rightarrow \bar{z} = \frac{-z}{z+1}$$

$$\Rightarrow z(z+1) = i \left(\frac{-z}{z+1} \right)$$

$$\Rightarrow (z+1)^2 = -i$$

$$\Rightarrow z+1 = \pm \left(e^{-i\pi/2} \right)^{1/2}$$

$$z+1 = \pm \frac{1-i}{\sqrt{2}}$$

$$z_1+1 = \frac{1-i}{\sqrt{2}}$$

$$z_2+1 = -\frac{1-i}{\sqrt{2}}$$

$$z_1+z_2 = -2$$

$$|z_1+z_2| = 2$$

10. Ans (8.00)

Let the variable chord be $x \cos \alpha + y \sin \alpha = p$.

which intersect the hyperbola

in A and B The joint equation of OA and OB

$$\text{is } \frac{x^2}{4} - \frac{y^2}{8} = \left(\frac{x \cos \alpha + y \sin \alpha}{p} \right)^2$$

$\Rightarrow$

$$\left(\frac{1}{4} - \frac{\cos^2 \alpha}{p^2} \right) x^2 - \left(\frac{1}{8} + \frac{\sin^2 \alpha}{p^2} \right) y^2 - \frac{2 \sin \alpha \cos \alpha}{p^2} xy = 0$$

$$\Rightarrow \frac{1}{4} - \frac{\cos^2 \alpha}{p^2} - \frac{1}{8} - \frac{\sin^2 \alpha}{p^2} = 0 \Rightarrow p^2 = 8$$

The variable line touches the fixed circle, thus

perpendicular distance of $(0, 0) = \text{Radius}$

$\therefore$ equation of the circle is $x^2 + y^2 = 8$.

